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Honors Thesis Proposal

Program: Cognitive Science

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After its release in November 2022, ChatGPT became the fastest-growing consumer application in history by reaching 100 million monthly active users in just two months. AI has become a pervasive part of modern life. 77% of the devices Americans use daily now incorporate AI features like voice assistants or chatbots¹, while in the corporate world, more than 85% of Fortune 500 companies have integrated AI into their operations². The healthcare sector, once with only 3% adoption, is now deploying AI at more than twice the rate of the broader economy³. This rapid adoption is especially alarming considering the inherent variability in output of current AI systems, the very existence of AI hallucinations, and the similarities within the processes we commonly ascribe to the language art of 'bullshitting'⁴. AI systems, specifically Large Language Models (LLMs, colloquially known as chatbots), are convenient and useful because they produce outputs that their creators knew we would be able to interpret, understand, and communicate with: natural, affective language. People can fall in love, feel heard, confide, and blindly follow a chatbot as their interlocutor. For some people, it takes years, even decades, to create a bond with a friend, a therapist, a love interest, a doctor, or even a parent. However, ChatGPT was first released only three years ago, in 2022; should we entrust chatbots to have such an intimate and varied relationship with our lives... only four years, or less, after we each individually met it? Help, aid, guidance, and what I'll best describe as the Irish-English word "craic" —a term for enjoyable, spirited conversation—are all successfully seductive traits that, in a plethora of cases, can use intimacy as exploitation. So, why do so many users – even those who might associate "seducers"⁵ with

1 [National University's Stats on AI Trends](#)

2 [Microsoft](#)

3 [AI in Healthcare](#)

4 As defined in Harry Frankfurt's "On Bullshit," a bullshitter is distinct from a liar. A liar knows the truth and seeks to obscure it; a bullshitter is indifferent to the truth, caring only about the persuasiveness of their statements. This maps closely to LLMs, which generate statistically plausible outputs without any underlying concept of truth or intent.

5 "seducer" is defined first as "A person who leads someone astray or entices someone into error or wrongdoing; a corrupter" in the [Oxford-English Dictionary](#). I personally think the idea of a "femme-fatale" in popular media is helpful to think of here.

its largely negative connotation – seem to readily confide in chatbot systems designed to interlocate, help, guide, and listen⁶?

This tendency is understandable. For many, communication with chatbots feels natural, and the incentives to trust them are high. They provide convenient, accessible, and often factually useful support for tasks from academic research to personal advice. But this trust is dangerous; reports of AI-induced “psychosis” and suicide, age-inappropriate advice, and hallucinations reveal a risk to our blind trust of chatbot outputs. The convenience of these systems creates an incentive to trust them, but *their* nature, as opposed to the anthropomorphized conversational nature we ascribe to them, makes that trust a hidden gamble.

This thesis will explore (and criticize) our relationship with artificial intelligence by laying out the fundamental conflict between **causal** and **intentional** explanations of the mind. To inform my readers of the conflict I find apparent, I will recount Descartes’ foundational discussions of the mind-body problem, Dennett and Searle’s stances on intentionality, Wittgenstein’s philosophical investigations which connect our use of language and meaning to intention, David Marr’s levels of analysis, and Pinker’s explanation of rationality as a cognitive and moral ideal. **Essentially, the final paper aims to prove a fundamental distinction: the human mind, as an intentional and meaning-making system, is of a different (or at least expanded) kind than the causal, mechanistic systems that produce AI outputs.**

To make this argument, I will explore how the human mind is not just different on the level of hardware⁷, but is a system whose biological origin enables it to be uniquely grounded in a physical and social world. It is this situated, affective, and intentional nature, along with the endured temporality of human life (like the aforementioned “years” it takes us to build trust), that distinguishes a human mind from a purely computational one. This is not to say that human minds don’t compute; it means they cannot be reduced to the purely causal systems of an LLM. I will distinguish this from ‘substance chauvinism’ by arguing that our biological implementation enables a different kind of cognition and computation—one

6 [Almost 40% of Americans Trust Medical Advice from Chatbots](#), [14.6% of Danish Students engage in friend like conversations with chatbots](#)

7 Hardware as it refers to the implementational level in Marr’s levels of analysis

inseparable from affect, intentionality, and meaning-making—that a causal system lacks. This will involve an analysis of Wittgenstein’s ‘shared life-forms’, Aristotelian and Simondonian accounts ofhylomorphism, and a refutation of the ‘Systems Reply’ to Searle’s Chinese Room.

My argument—that these two systems are fundamentally different—is motivated by a failure in our ‘folk psychology’; specifically, our inclination to attach a ‘mirage of understanding’ to chatbot outputs, a form of trust we often don’t afford to other humans⁸. I will use my argument, the consequence of distinguishing (and accepting the reality of both) causal and intentional architectures of mind, to explain and qualify our ability to believe we are understood by computational systems. I will then proceed to demystify this “understanding”. I am unsure of whether or not I will find what “computer understanding” is – whether it is a paradoxical term or an alien type of understanding – but, a foray into the history of computer science, information science, neuroscience, and taxonomy, and the naming of the entities and concepts in these fields will illuminate how we have come to be situated in a position where machine and human behavior become less distinguishable from one another.

Regardless of whether my understanding of the differences between humans and computers is correct, I believe that this investigation is worthwhile because it explains our ‘unnatural’ tendency to reverse a common cognitive process; Donald Davidson’s principle of charity suggests that humans naturally take each other seriously (assuming shared intentions) but not literally (we look past flawed words to find meaning). This is a product of our rationality and intuition for truth. Our engagement with chatbots shows that we reverse Davidson’s charity principle in the context of chatbots by interpreting AI’s mimicry of language literally, as an actual production of language. But we have not been taking AI seriously, a product of our folk psychology and the seductivity, convenience, and replication of natural language by a causal system.

To take AI seriously is to understand the causal origins and computational nature of its outputs, and in doing this, we will be better equipped to understand its power to alter our interpersonal

⁸ [In-group/out-group bias studies](#) show that we are often suspicious of others with a different social identity. This part of my proposal is an inquiry into why robotic/machine “identity” bypasses this bias

relationships, to complicate (or further) our understanding of the human condition, and to change our environment. In taking AI seriously, we allow ourselves to realize what is special about humans, the consequences of our technological creations, and the potential domestication and humanization of a distinct non-human intelligence. This understanding will enable me, as an engineer and consumer, to engage with AI without ignorantly or incorrectly equating its mind to my own.